

The changing sky

Our view into the universe is constantly changing. As Earth makes its daily spin, the Sun leaves the sky, darkness falls, and the distant stars come into view. Also, through the course of a year, the starry sky seen from a fixed point on Earth will change. The Sun, Moon, and planets follow their paths against the background of these stars. Particular star patterns—the zodiac constellations—help us locate them and track their progress.

■ YOUR CHANGING VIEW

One side of Earth is always facing the Sun, which lights up the daytime sky, while the other is facing the nighttime stars. Earth spins around on its axis, and as it makes its daily spin, an observer at a fixed location is facing toward both the Sun and the stars in any 24-hour period.

WHY THE VIEW ALTERS

If Earth did not move along its path around the Sun, it would alternately face the Sun and the same piece of starry sky. But because Earth travels around the Sun, it faces progressively different stars at the same time on successive nights during the course of a year. Although this change is hardly noticeable from night to night, as the weeks stretch into

months, the difference in the sky becomes more apparent. Observers in the northern hemisphere looking south and those in the southern hemisphere looking north will each see new constellations (see pp.12–13) with the changing seasons. By contrast, the northern sky for the northern-hemisphere observers and the southern sky for the southern-hemisphere observers undergo less change. These views contain the respective celestial poles, and the circumpolar stars that move around it are always in view.

As Earth makes its orbit, the starry backdrop to the Sun also changes. Like the Moon and planets, the Sun is seen against the zodiac band of sky (see facing page). Northern-hemisphere observers will see this stretching across the sky to their south, and southern observers see it in the sky to their north.



APRIL 1, 8 PM



APRIL 8, 8 PM

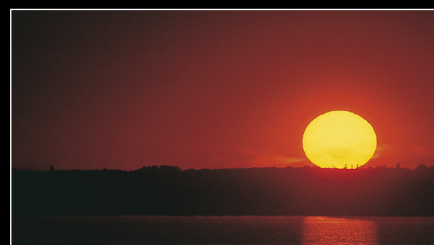
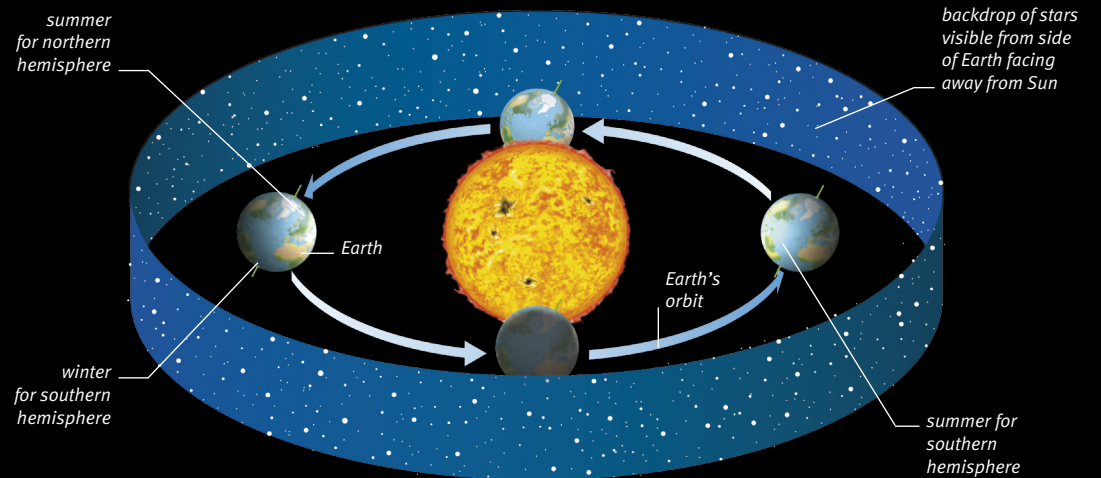


APRIL 15, 8 PM

Nighttime sky change

With the exception of those at the north and south poles, observers see different stars in their sky during the course of a year. An observer at 50°N sees the constellation of Orion lower and lower in the sky as April progresses. By the end of the month, it has almost disappeared below the horizon.

Seasonal sky change
One side of Earth faces the Sun, while the other side faces the stars. If Earth were fixed in space, it would face the Sun and the same piece of starry sky alternately as it makes its daily spin. But Earth is not fixed: it orbits the Sun. As a result, the stars we see in the summer months are different from those we see in winter.



AUGUST 6



AUGUST 7



AUGUST 11

Daytime sky change

As Earth makes its yearly orbit around the Sun, the position at which the Sun rises above the eastern horizon and sets below the western horizon changes. The three images above record sunset at a location 50°N on different days in August. The Sun sets farther and farther toward the south as the summer progresses.